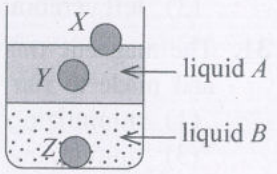
21. The main reason for the easy flow of electrons across the p-n junction of a forward biased diode is,
(1) high energy in the electrons.
(2) very low value of the potential barrier.
(3) that the depletion region is very narrow.
(4) that the doping level is very low.

22. There are two immiscible liquids A and B whose densities are ρ_A and ρ_B respectively. Three solid spheres X, Y and Z equal in radius and with densities ρ_X, ρ_Y and ρ_Z respectively are in the liquids as shown in the diagram. What is the correct statement about them?



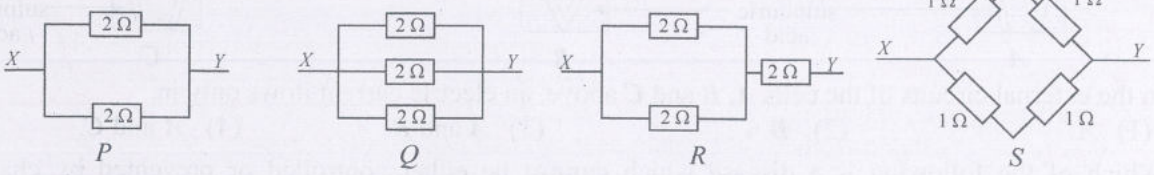
(1) $\rho_A < \rho_B$ and $\rho_X = \rho_Y < \rho_Z$
(2) $\rho_A < \rho_B$ and $\rho_X < \rho_Y < \rho_Z$
(3) $\rho_A = \rho_X = \rho_Y$ and $\rho_B < \rho_Z$
(4) $\rho_A = \rho_Y < \rho_X$ and $\rho_B = \rho_Z$

23. u, v and w are three metals lying above hydrogen in the activity series. The method proposed by a group of students to compare their activity was taking a piece from each metal, reacting them with hydrochloric acid separately and comparing the volumes of hydrogen gas liberated within a fixed short period. Which of the following factors is **not** essential to be kept constant during the experiment?
(1) surface area of the metals
(2) mass of the metals
(3) temperature of the reaction mixture
(4) acid concentration of the reaction mixture

24. During inspiration,
(1) intercostal muscles and the muscles in the diaphragm relax.
(2) intercostal muscles relax while muscles in the diaphragm contract.
(3) intercostal muscles contract while muscles in the diaphragm relax.
(4) intercostal muscles and muscles in the diaphragm contract.

25. Consider the following statements about amoeba.
A - Reproduction occurs by binary fission.
B - Binary fission occurs only by mitotic division.
Of the above statements,
(1) only statement A is true.
(2) only statement B is true.
(3) both statements A and B are true.
(4) both statements A and B are false.

26. Of P, Q, R and S, in which pair of resistor systems is the value of the equivalent resistance the same between X and Y?



(1) P and Q
(2) P and S
(3) Q and R
(4) R and S

27. Two metals Q and R react as indicated by the following equations.

$Q(s) + H_2SO_4(aq) \longrightarrow QSO_4(aq) + H_2(g)$

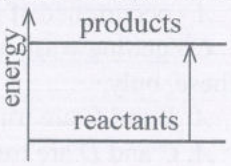
$R(s) + H_2SO_4(aq) \longrightarrow RSO_4(aq) + H_2(g)$

$R(s) + QSO_4(aq) \longrightarrow$ no reaction

Which of the following can be Q and R respectively?
(1) Cu and Zn
(2) Zn and Mg
(3) Mg and Cu
(4) Mg and Zn

28. A gas jar filled with a gas X was inverted and placed vertically in a beaker of water. Then, it was seen that the water level inside the gas jar rises rapidly. Gas X can be,
(1) Ne.
(2) N_2.
(3) CO_2.
(4) NH_3.

29. Which of the following is most likely to be the change represented by the energy level diagram given?
(1) dissolving of glucose in water
(2) combustion of the hydrocarbon, methane
(3) reaction between zinc and hydrochloric acid
(4) neutralization between sodium hydroxide and hydrochloric acid



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